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I've been working on this research paper for several months now. What are its fatal weaknesses ? how can i complete it into the platonic ideal shape it yearns to be, guided by its own inner music? what are the top 1000 literal line item fixes I have to make, that altogether UPGRADE the mathematics and the physics? never downgrade? Only tolerate realizing the strongest possible version of this paper imaginable/conceivable.

I need to get this to a publishable shape soon, some time this year (can be end of year) so I can put the preprint on arxiv. Recall that every line of this manuscript needs to be such that one of witten, dirac, etingof, gelfand, bezrukavnikov, polyakov, gaiotto, costello could have written it.

your work is typically downgrading. what mathematics and technical detail is missing to actually prove the strongest form of the conjectures contained within? what missing technical details are needed to construct the strongest form of the mathematics? the greatest algebraic / homotopical / analytical edifice? the most tangible, concrete, mathematical structures? Your outputs should not be project management, or coding scaffolding, or anything except ACTUAL PURE MATHEMATICS: algebra, geometry, homotopy theory

Show less ^

Thought for 10m 15s >

I read the uploaded manuscript as a theorem/proof object, not as prose. Its strongest mathematical core is the already-known Borchers–Gritsenko–Nikulin triangle:

$$D_X = \Delta_5, \quad \text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z), \quad \Theta_{\text{Bor}}(\phi_{0,1}^{\text{Weil}}) = \Delta_5.$$

That triangle is real. The manuscript is strongest wherever it stays inside it. Its fatal weakness is the attempted passage

$$\text{Borchers/BKM/theta data} \rightsquigarrow K3 \times E \text{ compact source} \rightsquigarrow \text{Pf}^{\text{prot}}(\mathfrak{D}_X^{DI}) = \Delta_5 \rightsquigarrow P_X \simeq \mathfrak{g}_{\Delta_5},$$

because the paper mostly **names** the compact-source, orientation, Hall, Koszul, Pfaffian, and primitive-recognition data rather than constructing them. Your own manuscript says the automorphic/BKM/theta edges are theorems, the Pfaffian edge is conditional on $(D0), (O1), (O1)^+, (O2_{\text{atlas}}), (P_{\text{fin}})$, and primitive recognition still requires source representatives, relations, pairings, radical quotient, PBW comparison, and full parity dimensions. main The construction theorem for the Dirac–Igusa package is also explicitly “granted” the retained D0–HN datum, quotient-orientation datum, Weyl lifts, $O2_{\text{atlas}}$, finite-stage hypotheses, E_3 –HolFA, $K3 \rightarrow E$ specialization, chiral Koszul datum, ML exactness, and cofinal primitive-recognition datum. main

The paper’s inner music is not “five views of Δ_5 .” That is the expository surface. Its real inner music is the following theorem-schema:

Construct a reduced, oriented, hybrid, chiral Hall source on $K3 \times E$ whose primitive Lie superalgebra is the GN

Everything else should be subordinate to that.

Fatal weaknesses

- 1. The main theorem is presently conditional in almost every nonclassical place

The paper repeatedly says “retained datum,” “chosen datum,” “standing hypothesis,” “finite-stage input,” and “obstruction tuple vanishes.” That is not fatal for a program paper, but it is fatal for a theorem paper. The Dirac–Igusa object is defined as an inverse limit of finite constituents, yet each constituent carries its own obstruction class and is not inferred from the others. ^{main} The paper’s current logical form is closer to:

If the desired compact Hall/Pfaffian/Koszul/recognition package exists, then it realizes Δ_5 .

The publishable theorem must instead construct enough of that package that the implication has non-tautological content.

2. Primitive recognition is not proved; the manuscript itself gives the counterexample

The manuscript correctly says that signed superdimensions do not recognize $\mathfrak{g}_{\Delta_5}$: a zero-bracket graded super vector space with the same signed dimensions has the same denominator but is not the BKM Lie superalgebra. ^{main} Therefore the equality

$$\text{sdim } P_{X,\gamma}^{\Pi,\text{red}} = f(nm, l)$$

is almost irrelevant unless accompanied by the actual bracket, Chevalley relations, real Serre relations, Borchers orthogonality, Hopf radical, kernel equality, PBW comparison, and full parity dimensions. Your manuscript lists exactly these as $(S1)–(S10)$. ^{main}

The first real proof target is not Δ_5 . It is:

$$\ker(\mathfrak{F}(B_{\Delta_5}) \rightarrow P_X^{\Pi,\text{red}}) = J_{\text{BK}} + \text{Rad}_{\text{GN}}$$

on a cofinal system of relation-closed finite windows, with strict PBW and $\lim^1 = 0$.

3. The Hall bialgebra is not built on $K\mathfrak{z} \times E$

The manuscript knows that the Hall coproduct requires extra six-functor hypotheses, because the reversed extension correspondence is not automatically proper in the compact $K\mathfrak{z} \times E$ problem. It also says compatibility of product and coproduct is a retained bialgebra residual, not a consequence of the product theorem. ^{main}

This is a fatal gap because primitives are defined by

$$\text{Prim}(H) = \ker(\Delta - 1 \otimes \text{id} - \text{id} \otimes 1),$$

so without an actual coproduct and Hopf pairing, the primitive algebra P_X is not a constructed object.

4. The orientation theorem is still a list of gerbes and characters, not a computation

The paper’s orientation lane requires vanishing and compatible null-trivialization of reduced and equivariant Čech–Borel classes, plus Coxeter coherence and torsor defect vanishing for Weyl transport.

^{main} To be a theorem, this must become an actual calculation of

$$w_2 \det R\text{Hom}^{\text{red}}(F, F), \quad \beta_{C,S}^H \in H^2(BH; \mathbb{F}_2), \quad \lambda_{C,S}^H \in H^1(BH; \mathbb{F}_2),$$

for every retained stabilizer stratum. Right now the notation is much stronger than the proof.

5. The Koszul comparison is assumed at exactly the place where it must be proved

The paper separates source and target correctly: the target counit of the imported current envelope does not create compact source data, and the cobar map

$$\Theta_{\text{Kos},R} : \Omega_E^{\text{ch}} C_X \rightarrow \text{Den}_{\Delta_5,E}$$

is a source-to-target comparison. ^{main} But the actual quasi-isomorphism is conditional on the chiral Koszul source datum and Koszul Mittag–Leffler exactness; the wrapped Koszul term, the source-to-target map, and ML exactness remain conditions. ^{main}

This is not a small gap. It is the homotopical heart of the paper.

6. The Pfaffian is not yet an analytic/geometric determinant

The manuscript identifies the protected Pfaffian with Δ_5 only relative to retained data. The scalar branch $-4096\Delta_5^{-2}$ is distinguished from the orientation character, and the text explicitly says no compact microscopic state space is asserted. main

The missing theorem is not “the product equals Δ_5 .” The missing theorem is:

$$\mathrm{Pf}\left(D_{\mathrm{hyb},R}^{\mathrm{red}}\right)=64q^{1/2}r^{1/2}s^{1/2}\prod_{\Gamma_{\mathrm{eff}}}(1-q^nr^ls^m)^{f(nm,l)}$$

as a section of a constructed Pfaffian line, with its wall-crossing orientation character computed.

7. The mirror discriminant is correctly marked conjectural, but it should not be entangled with the proof

The manuscript says the mirror-discriminant identification requires existence of a rank-3 Mukai–Hodge quotient and comparison of the automorphic line with the period Hodge line; the automorphic theorem gives the Humbert divisor, but equality with the mirror discriminant is conjectural. main This is not a weakness if kept separate. It becomes fatal only if the reader feels it is used to support the Pfaffian/Hall theorem.

8. The scalar trace is overused

The scalar identity

$$Z_{K3\times E}^{\mathrm{BPS}}=\Delta_5^{-2}$$

forgets the orientation character. The manuscript itself says the operator-level datum carries ϵ_o , while the scalar shadow squares it away. main Any attempt to infer the Pfaffian line, the primitive bracket, or the compact state space from Δ_5^{-2} is mathematically invalid.

The strongest possible mathematical shape

The paper wants to become the following theorem.

Let $X=S\times E$, S a smooth projective K3 surface and E an elliptic curve. Let

$$\Gamma_X=\Gamma_X^{\mathrm{form}}\oplus_B\Gamma_{\mathrm{gram}},\qquad\Pi_X(c,T)=\Pi_X(c)-T,$$

where $B=-\delta\Pi_X$ is the Mukai–Gram polarization cocycle. Let R range through a cofinal HN system. Then one should construct:

$$\mathcal{F}_{X,R}^{\mathrm{hyb}}\in\mathrm{Fact}_{E,\mathrm{hyb}}^{E_3,\mathrm{red}}$$

with orientation o_R , vanishing-cycle coefficients Φ_R , Hall product m_R , Hall coproduct Δ_R , Hopf pairing $\langle-,-\rangle_R$, Pfaffian line $L_{\mathrm{Pf},R}$, Pfaffian section $\mathrm{pf}_{X,R}$, chiral source coalgebra $C_{X,R}$, and a normal-ordered primitive Lie superalgebra

$$P_{X,R}^{\mathrm{II},\mathrm{red}}=\mathrm{Prim}\left(H_{\bullet}^{\mathrm{BM}}(\mathfrak{M}_{X,R}^{\mathrm{red},+},\Phi_R\otimes o_R)\right)/\mathrm{Rad}\langle-,-\rangle_R.$$

Then prove the simultaneous identities:

$$\lim_R\mathrm{Pf}^{\mathrm{prot}}(\mathfrak{D}_{X,R}^{DI})=\Delta_5,$$

$$\epsilon_o=\nu_{\Delta_5}\quad\text{on }W^{(2)}(\Lambda_{II}^{2,1}),$$

$$\lim_RP_{X,R}^{\mathrm{II},\mathrm{red}}\cong\mathfrak{g}_{\Delta_5},$$

$$\Omega_E^{\mathrm{ch}}C_X\simeq U_E^{\mathrm{ch}}(\mathrm{Cur}_E\mathfrak{g}_{\Delta_5}).$$

The proof should run through a finite-window spectral sequence:

$$E_{\beta,p,q}^1=\bigoplus_{\substack{\beta=\beta_1+\cdots+\beta_k\\ \beta_i\text{ simple}}}H_q^{\mathrm{BM}}(\mathfrak{gl}\mathfrak{a}\mathfrak{g}_{\beta_1,\dots,\beta_k}^{\mathrm{red}},\Phi_{\beta_1,\dots,\beta_k}\otimes o)\Rightarrow P_{X,\beta}^{\mathrm{II},\mathrm{red}},$$

whose d_1 is the Hall boundary, whose E_2 quotient is the Borchers–Kac presentation, and whose E_∞ has exactly the GN parity dimensions. That is the actual algebraic/homotopical edifice the paper needs.

Top 1000 literal mathematical fixes

I am writing these as missing mathematical artifacts, not as writing advice.

A. Logical status and theorem architecture

1. Replace every unconditional $P_X \simeq \mathfrak{g}_{\Delta_5}$ by a theorem with $(S1)–(S10)$ hypotheses.
2. Prove at least one nontrivial finite-window instance of $P_X \simeq \mathfrak{g}_{\Delta_5}$.
3. Separate theorem, conjecture, conditional theorem, retained datum, and definition typographically.
4. State the exact category in which $\mathfrak{D}_X^{DI}$ lives.
5. State the exact topology on the inverse limit over R .
6. Define the finite HN index category R_{HN} as a filtered category.
7. Prove transition maps in R_{HN} are strict on filtrations.
8. Prove $\varprojlim^1 = 0$ for each inverse system used.
9. Give a single master diagram of all finite-stage maps.
10. Define “retained datum” as a mathematical object, not prose.
11. Replace “chosen Weyl lifts” by an explicit torsor of lifts.
12. Replace “retained quotient-orientation datum” by a cocycle complex.
13. Replace “finite Pfaffian section datum” by a section over a line bundle.
14. Replace “hybrid source datum” by a colored factorization category.
15. Replace “recognition datum” by source bases and matrices.
16. Give a dependency graph of objects as morphisms in categories.
17. Prove no theorem uses the mirror conjecture.
18. Prove no theorem uses the gravity-line residual.
19. Prove no theorem uses the scalar trace to recover orientation.
20. Define the Beilinson chain functorially.
21. Define levels P, C, S, Z, A as categories or strata.
22. State which equalities live in K_0 .
23. State which equalities live in Picard groupoids.
24. State which equalities live in derived categories.
25. State which equalities live in completed Lie superalgebras.
26. State which equalities live in automorphic line bundles.
27. State which equalities live only after scalar trace.
28. Prove $D_X = \Delta_5$ is not used to define Pf^{prot} .
29. Prove Pf^{prot} is not defined by its desired value.
30. Prove P_X is not defined by target multiplicities.
31. Give a universal property of $\mathfrak{D}_X^{DI}$.
32. Prove $\mathfrak{D}_X^{DI}$ is independent of finite presentation choices.
33. Define morphisms between Dirac–Igusa finite stages.
34. Define isomorphism of Dirac–Igusa pro-objects.
35. Prove finite-stage objects form a pro-category.
36. Give base-change functoriality in S .
37. Give base-change functoriality in E .
38. Define dependence on the K3 polarization.
39. Define dependence on elliptic origin 0_E .
40. Prove independence of 0_E up to canonical character.
41. State the exact theorem obtained without primitive recognition.
42. State the exact theorem obtained without Koszul comparison.
43. State the exact theorem obtained without Pfaffian section.
44. State the exact theorem obtained without $O2_{\text{atlas}}$.
45. State the exact theorem obtained without $O1^+$.
46. State the exact theorem obtained without $O1$.
47. State the exact theorem obtained without $D0$.

48. Prove each implication between obstruction vanishings.
49. Prove no obstruction vanishing is redundant.
50. Give counterexamples showing each obstruction is necessary.

B. Automorphic, Jacobi, and BKM arithmetic

51. Fix one normalization of $\phi_{0,1}$ globally.
52. Record $\phi_{0,1} = r^{-1} + 10 + r + O(q)$ once and propagate it.
53. Prove $f(n, l) = f(n, -l)$.
54. Prove the discriminant dependence $f(n, l) = c(4n - l^2)$.
55. Tabulate $f(n, l)$ through the first relation-closed window.
56. Derive the product for $D_5 = 64^{-1}\Delta_5$.
57. Derive the theta-leading coefficient 64.
58. Prove $\Delta_5^2 = \chi_{10}$ in the chosen normalization.
59. Prove $\chi_{10}^{OP} = 4096^{-1}\Delta_5^2$.
60. Keep $\Delta_{10}, \chi_{10}, \chi_{10}^{OP}, \Phi_{10}^{un}$ distinct.
61. Define ν_{Δ_5} as a character, not an orientation.
62. Compute $\nu_{\Delta_5}(s_{\delta_i}) = -1$.
63. Compute ν_{Δ_5} on type-I chamber automorphisms.
64. Prove $\nu_{\Delta_5}|_{W^{(2)}} = \det$.
65. Prove $\nu_{\Delta_5}^2 = 1$.
66. Define the automorphic line $L^5 \otimes \nu_{\Delta_5}$.
67. Prove $\text{div } \Delta_5 = H_{\text{Hum}}$.
68. Prove the divisor multiplicity is one.
69. Prove $\text{div } \Delta_5^{-2} = 2H_{\text{Hum}}$.
70. Separate divisor support from divisor multiplicity.
71. Derive the singular theta lift from the Weil representation.
72. Define $\phi_{0,1}^{\text{Weil}}$ explicitly.
73. Write the two theta components h_0, h_1 .
74. Compute the T -matrix on (h_0, h_1) .
75. Compute the S -matrix on (h_0, h_1) .
76. Identify the discriminant form of $\Lambda^{3,2}$.
77. Prove the hyperbolic summands do not change the discriminant form.
78. State the regularized theta integral with the correct v -factor.
79. State the Weyl vector formula from the principal part.
80. Derive $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$.
81. Prove $(\rho, \delta_i) = -1$.
82. Prove $(\rho, \rho) = -3/2$.
83. Prove $\rho \in (\Lambda_{II}^{2,1})^\vee$.
84. Define the product chamber $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$.
85. Prove the semigroup Γ_{eff} is closed under needed decompositions.
86. Define Γ_{act} .
87. Prove inactive triples contribute exponent zero.
88. Prove $Z \mapsto 2Z$ corresponds to the type-II root lattice.
89. Prove the root map $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$.
90. Prove $\alpha(\Gamma_{\text{eff}}) \subset \Lambda_{II}^{2,1}$.
91. Prove $\alpha(n, l, m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3$.
92. Prove $n + m - l \geq 0$ on active support.
93. Prove $C_{\text{Borch}} = \mathbb{R}_{\geq 0}\langle \delta_1, \delta_2, \delta_3 \rangle$.
94. Define $\Lambda^{2,1}$ with a single sign convention.
95. Define $\Lambda_{II}^{2,1} \cong U(4) \oplus \langle 2 \rangle$.
96. Compute its discriminant -32 .
97. Compute $(\Lambda_{II}^{2,1})^\vee$.
98. Prove the three δ_i are primitive type-II square-two roots.
99. Compute the Cartan matrix $4I_3 - 2J_3$.
100. Prove $W^{(2)}(\Lambda_{II}^{2,1})$ is generated by the three s_{δ_i} .
101. State whether any Coxeter braid relations are finite.

102. Define $\text{Aut}(P_{II}) \cong S_3$.
103. Compute its action on δ_i .
104. Prove ρ is S_3 -fixed.
105. Define type-I versus type-II root divisibility.
106. Prove type-I walls are absent from $\text{div } \Delta_5$.
107. Prove the affine $A_1^{(1)}$ subalgebras on δ_i, δ_j .
108. Compute isotropic roots $a_{ij} = \delta_i + \delta_j$.
109. Prove $\text{smult}(a_{ij}) = 10$.
110. Decompose $10 = 1 + 9$ at a_{ij} .
111. Compute the GN isotropic simple directions $u_{ij,r}$.
112. Prove $\tau(ta_{ij}) = 9$.
113. Compute $\delta_{123} = \delta_1 + \delta_2 + \delta_3$.
114. Compute $(\delta_{123}, \delta_{123}) = -6$.
115. Compute $\text{smult}(\delta_{123}) = -64$.
116. Prove the target parity split $29|93$.
117. Derive the two real-real-real directions.
118. Derive the 27 real-isotropic mixed directions.
119. Derive the 93 odd imaginary simple directions.
120. Compute all target parity data through the first relation-closed window.

C. Mukai–Gram and normal ordering

121. Define $\Gamma_X^{\text{form}} = H(S, \mathbb{Z}) \oplus H(S, \mathbb{Z})$.
122. Define Γ_X^{alg} .
123. Define $\Gamma_{X,\sigma}^{\text{eff}}$.
124. Prove $\Gamma_{X,\sigma}^{\text{eff}} \subset \Gamma_X^{\text{alg}} \subset \Gamma_X^{\text{form}}$.
125. Define the Mukai pairing with fixed sign.
126. Prove $H(S, \mathbb{Z}) \cong II_{4,20}$.
127. Define Q, P components under Künneth.
128. Define $\Pi_X(Q, P) = ((Q, Q)/2, (Q, P), (P, P)/2)$.
129. Prove Π_X is integer-valued.
130. Compute the polarization cocycle $B(c, c')$.
131. Prove $B = -\delta\Pi_X$.
132. Prove $B(c, c) = 2\Pi_X(c)$.
133. Prove B is a normalized group 2-cocycle.
134. Define $\Gamma_X = \Gamma_X^{\text{form}} \oplus_B \Gamma_{\text{gram}}$.
135. Prove Γ_X is abelian.
136. Prove $\Pi_X : \Gamma_X \rightarrow \Gamma_{\text{gram}}$ is additive.
137. Prove the split section $s(c) = (c, \Pi_X(c))$ is additive.
138. Prove fixed-lift raw pushforward fails on second real-root strings.
139. Show the failure explicitly for $2\delta_i + \delta_j$.
140. Prove $B(c_i, c_i + c_j) = 2\Pi_X(c_i)$.
141. Prove no linear primitive trivializes B .
142. Define mixed-lift fibre-summed pushforward.
143. Decide whether mixed-lift pushforward can support brackets.
144. Prove normal ordering is functorial under Hall addition.
145. Prove normal ordering is functorial under coproduct.
146. Prove normal ordering respects primitive projection.
147. Prove normal ordering respects orientation twists.
148. Prove normal ordering respects Weyl reflection lifts.
149. Prove $\alpha \circ \Pi_X$ is the root grading.
150. Prove source degrees with same Gram label are correctly summed.
151. Define the kernel of Π_X .
152. Prove the kernel does not create hidden zero-superdimension primitives.
153. Define the radical along $\ker \Pi_X$.
154. Prove $\Pi_{X,*}^{\Theta}$ is exact on finite windows.
155. Prove $\Pi_{X,*}^{\Theta}$ commutes with bar construction.

- 156. Prove $\Pi_{X,*}^{\Theta}$ commutes with Hall bracket.
- 157. Prove $\Pi_{X,*}^{\Theta}$ commutes with Hopf pairing.
- 158. Prove $\Pi_{X,*}^{\Theta}$ preserves parity.
- 159. Prove $\Pi_{X,*}^{\Theta}$ preserves PBW filtration.
- 160. Prove $\Pi_{X,*}^{\Theta}$ preserves triangular decomposition.

D. Stability, finite moduli, and HN windows

- 161. Define the chosen stability heart on $D^b \text{Coh}(K3 \times E)$.
- 162. Prove it is noetherian on retained windows.
- 163. Prove Harder–Narasimhan filtrations exist.
- 164. Prove HN type is bounded in each active window.
- 165. Define Liu product–stability precisely.
- 166. Prove compatibility with Abramovich–Polishchuk heart.
- 167. Prove finite-type semistable substacks for each retained class.
- 168. Prove quasi-smooth derived Artin enhancement.
- 169. Prove finite residual inertia after rigidification.
- 170. Define the finite class set C_R .
- 171. Define Hilbert polynomial bounds $P_{c,i}$.
- 172. Define cohomological amplitude bounds $[a_c, b_c]$.
- 173. Prove finite regularity bound N_c .
- 174. Construct universal perfect complexes on finite substacks.
- 175. Prove scalar rigidification exists.
- 176. Prove E -translation rigidification exists.
- 177. Define the retained closed substacks.
- 178. Prove retained substacks are closed under extensions.
- 179. Prove retained substacks are closed under HN factors.
- 180. Prove retained substacks are closed under duals if needed.
- 181. Define downward-saturated root windows W_{ν} .
- 182. Define $\Gamma(W_{\nu}) = \{\gamma : \alpha(\gamma) \in W_{\nu}\}$.
- 183. Prove active windows exhaust R^+ .
- 184. Prove source windows exhaust compact support.
- 185. Prove target windows and source windows are compatible.
- 186. Prove finite moduli transitions are closed embeddings or proper maps.
- 187. Prove transition maps preserve orientations.
- 188. Prove transition maps preserve vanishing cycles.
- 189. Prove transition maps preserve Hall products.
- 190. Prove transition maps preserve Hall coproducts.
- 191. Prove transition maps preserve primitive subspaces.
- 192. Prove transition maps preserve radicals.
- 193. Prove transition maps preserve PBW filtrations.
- 194. Prove transition maps preserve parity decompositions.
- 195. Prove $\lim^1$ vanishes on moduli cohomology.
- 196. Prove $\lim^1$ vanishes on primitive spaces.
- 197. Prove $\lim^1$ vanishes on pairing kernels.
- 198. Prove $\lim^1$ vanishes on PBW associated gradeds.
- 199. Define the finite test window Γ_R^{test} .
- 200. Prove every relation-closed target degree appears at finite R .

E. Hybrid Ran carrier and wrapped geometry

- 201. Define b_R^{geom} as an actual degree map.
- 202. Prove b_R^{geom} is additive under retained extensions.
- 203. Distinguish b_R^{geom} from $m = \text{pr}_3 \Pi_X$.
- 204. Prove positive elliptic degree projects onto all of E .
- 205. Define $\text{Ran}^{\text{hyb}}(E)$ as a stack or prestack.
- 206. Define its local stratum $b = 0$.
- 207. Define its wrapped stratum $b > 0$.

208. Define mixed local/wrapped correspondences.
209. Define wrapped/wrapped correspondences.
210. Define local/local correspondences.
211. Define ordered two-sided mixed correspondences.
212. Define source and target anchors before quotient.
213. Construct the determinant anchor $\lambda(F)$.
214. Compute its translation weight.
215. Handle $\chi(F) = 0$ anchor degeneracy.
216. Add extra anchor data distinguishing $O_E^{\oplus 2}$ from $L \oplus L^{-1}$.
217. Define anchor residual $\mathfrak{o}_R^\lambda$.
218. Prove anchor residual vanishes in first window.
219. Define E -equivariant wrapped prequotient stacks.
220. Prove finite type for wrapped prequotients.
221. Prove properness of local/local extension stacks.
222. Prove admissibility of mixed extension stacks.
223. Prove admissibility of wrapped/wrapped extension stacks.
224. Define compact-support exceptional pushforward model.
225. Prove base-change for mixed correspondences.
226. Prove projection formula for mixed correspondences.
227. Prove Thom–Sebastiani transport for mixed correspondences.
228. Define the eight words $LLL, \dots, WWW$.
229. Construct the eight two-step flag stacks.
230. Prove associativity for LLL .
231. Prove associativity for LLW .
232. Prove associativity for LWL .
233. Prove associativity for WLL .
234. Prove associativity for LWW .
235. Prove associativity for WLW .
236. Prove associativity for WWL .
237. Prove associativity for WWW .
238. Prove pentagon coherence for four inputs.
239. Define the higher-colored tree category.
240. Prove higher-colored residual $\mathfrak{o}_R^{\text{col}} = 0$.
241. Define units in the hybrid factorization structure.
242. Prove unit compatibility with local inputs.
243. Prove unit compatibility with wrapped inputs.
244. Prove symmetric-group descent on local configurations.
245. Prove braid/order conventions for wrapped insertions.
246. Define quotient-after-correspondence pseudofunctor.
247. Prove quotient-after-correspondence preserves composition.
248. Prove quotient-after-correspondence preserves base change.
249. Prove quotient-after-correspondence preserves Thom–Sebastiani.
250. Prove quotient-first construction is not being used.
251. Construct $Q_{E,R}$ on Borel–Moore chains.
252. Construct comparison maps $\theta_{\mu,R}^Q$.
253. Prove $\theta_{\mu,R}^Q$ is compatible with associativity.
254. Prove $\theta_{\mu,R}^Q$ is compatible with coproduct.
255. Prove $\theta_{\mu,R}^Q$ is compatible with primitives.
256. Prove HN transitions commute with $Q_{E,R}$.
257. Define protected integration I_R^{prot} .
258. Prove I_R^{prot} respects product.
259. Prove I_R^{prot} respects coproduct.
260. Prove I_R^{prot} respects primitive projection.
261. Prove I_R^{prot} has Gram degree s^m .
262. Prove I_R^{prot} uses Π_X , not geometric b .
263. Compute images of $\gamma(\delta_1) = (1, 1, 0)$.

- 264. Compute images of $\gamma(\delta_2) = (0, 1, 1)$.
- 265. Compute images of $\gamma(\delta_3) = (0, -1, 0)$.
- 266. Construct compact representatives for the three wall charges.
- 267. Prove the middle wall $(0, 1, 1)$ lies in wrapped/mixed geometry.
- 268. Prove hybrid operations recover the three type-II walls.
- 269. Prove no type-II wall is lost under E -quotient.
- 270. Prove no false type-I wall enters the hybrid carrier.

F. Reduced d-critical orientation

- 271. Construct the PTVV (-1) -shifted symplectic form on finite substacks.
- 272. Construct the Joyce d-critical truncation.
- 273. Construct the cosection from the K3 holomorphic 2-form.
- 274. Prove cosection surjectivity on retained strata.
- 275. Define $R\mathrm{Hom}^{\mathrm{red}}(F, F)$.
- 276. Prove the reduced obstruction complex is perfect.
- 277. Define $\det R\mathrm{Hom}^{\mathrm{red}}(F, F)$.
- 278. Construct its square root $\mathcal{O}_F$.
- 279. Compute w_2 of the determinant line.
- 280. Prove $\alpha_C^{\mathrm{red}} = 0$.
- 281. Construct a null-homotopy of α_C^{red} .
- 282. Compute the free E -Borel class.
- 283. Prove $\alpha_C^{E, \mathrm{free}} = 0$.
- 284. Construct its null-trivialization.
- 285. Compute finite stabilizer class $\beta_{C,S}^H$.
- 286. Prove $\beta_{C,S}^H = 0$ for every retained H .
- 287. Construct null-trivializations compatible under subgroup restriction.
- 288. Compute linearization character $\lambda_{C,S}^H$.
- 289. Prove $\lambda_{C,S}^H = 0$.
- 290. Prove gerbe vanishing does not imply linearization vanishing.
- 291. Compute the $E[2]$ Klein-four class.
- 292. Derive $b_{20} = r_1$.
- 293. Derive $b_{11} = r_1 + r_2 + r_3$.
- 294. Derive $b_{02} = r_2$.
- 295. Prove $r_1 = r_2 = r_3 = 0$.
- 296. Compute even $N \geq 4$ two-primary classes.
- 297. Detect the mixed $x_1 x_2$ term.
- 298. Prove odd N transfer kills mod-2 obstruction.
- 299. Prove orientation multiplicativity under direct sum.
- 300. Prove orientation multiplicativity under extensions.
- 301. Prove Thom–Sebastiani compatibility for orientation lines.
- 302. Prove orientation descent through E -quotient.
- 303. Prove orientation descent commutes with Hall product.
- 304. Prove orientation descent commutes with Hall coproduct.
- 305. Prove orientation descent commutes with primitive projection.
- 306. Construct Weyl wall transport $\tau_i : \mathcal{O} \rightarrow s_{\delta_i}^* \mathcal{O}$.
- 307. Prove $\tau_i^2 = 1$.
- 308. Define the finite partial action groupoid G_R .
- 309. Define the orientation 2-cocycle $c_{\mathcal{O}, R}$.
- 310. Prove $[c_{\mathcal{O}, R}] = 0$.
- 311. Construct a 1-cochain killing $c_{\mathcal{O}, R}$.
- 312. Prove compatibility under $R \rightarrow R'$.
- 313. Compute torsor defects $\omega_{i,C}$.
- 314. Prove all $\omega_{i,C} = 0$.
- 315. Prove Weyl transport preserves α^{red} .
- 316. Prove Weyl transport preserves $\alpha^{E, \mathrm{free}}$.
- 317. Prove Weyl transport preserves finite stabilizer β^H .

318. Prove Weyl transport preserves $\lambda^H = 0$.
319. Define the orientation character ϵ_o .
320. Compute $\epsilon_o(s_{\delta_i}) = -1$.
321. Prove $\epsilon_o = \det$ on $W^{(2)}$.
322. Prove $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}$.
323. Prove S_3 chamber automorphisms have trivial Maass character.
324. Decide whether ϵ_o extends to $W^{(2)} \rtimes S_3$.
325. Construct semidirect Hall lifts if extension is claimed.
326. Prove no imaginary divisor monodromy is being called Weyl.
327. Define local divisor monodromy separately.
328. Prove local divisor monodromy is irrelevant to ν_{Δ_5} .
329. Compute orientation on the three simple wall charts.
330. Compute orientation on isotropic a_{ij} charts.
331. Compute orientation on δ_{123} chart.
332. Prove orientation line is compatible with parity split 29|93.
333. Prove orientation line is compatible with negative roots.
334. Construct Chevalley anti-involution on orientation data.
335. Prove orientation pairing is Frobenius.
336. Prove Frobenius trace has degree -3 .
337. Prove signs agree with CY3 Serre duality.
338. Prove reduced CY3 pairing is nondegenerate modulo radical.
339. Prove orientation classes survive inverse limit.
340. Prove no new orientation class appears at the limit.

G. Hall product, coproduct, pairing, and primitives

341. Define the finite Hall object H_R^{geom} .
342. Define $m_R = q! \text{TS}^{\text{red}} p^*$.
343. Prove p^* exists in the chosen six-functor formalism.
344. Prove $q!$ exists for retained correspondences.
345. Prove $q!$ is independent of compactification.
346. Prove associativity of m_R via two-step flags.
347. Define $\Delta_R = p! \text{TS}^{-1} q^*$.
348. Prove $p!$ exists for reversed correspondences.
349. Prove coassociativity of Δ_R .
350. Prove bialgebra compatibility $\Delta m = (m \otimes m) \Delta$.
351. Define the Hall antipode if a Hopf algebra is claimed.
352. If no antipode, say bialgebra not Hopf.
353. Define the primitive subspace $P_R = \ker \bar{\Delta}_R$.
354. Prove the commutator of primitives is primitive.
355. Prove the Hall bracket satisfies graded Jacobi.
356. Prove the bracket preserves normal-ordered Gram degree.
357. Prove the bracket respects parity.
358. Define the positive-negative Hall pairing.
359. Prove Hall pairing is homogeneous.
360. Prove Hall pairing is supersymmetric in the required sense.
361. Prove Hall pairing is invariant: $\langle [x, y], z \rangle = \langle x, [y, z] \rangle$.
362. Prove Hall pairing is compatible with coproduct.
363. Define the radical K_γ .
364. Prove K_γ is a Lie ideal.
365. Prove K_γ is a coideal.
366. Prove the radical quotient is again a bialgebra.
367. Prove radical quotient is finite-dimensional on windows.
368. Define pairing matrices $G_{\gamma, ab}$.
369. Compute G_{δ_i} .
370. Compute $G_{a_{ij}}$.
371. Compute $G_{\delta_{123}}$.

372. Compute kernels in the first arithmetic window.
373. Compute kernels in the first relation-closed window.
374. Prove kernels agree with GN/Kac kernels.
375. Define source quotient matrices Q_γ .
376. Prove $Q_\gamma G_\gamma Q_\gamma^T$ is nondegenerate.
377. Prove compatibility with finite transitions.
378. Define the integral form of the Hall algebra.
379. Prove the integral form is stable under product.
380. Prove the integral form is stable under coproduct.
381. Prove the integral form is stable under pairing.
382. Define completion for $Y^+(X)$.
383. Prove completion is separated.
384. Prove completion is complete.
385. Define the Drinfeld double $D(Y^+(X))$.
386. Prove cross-relations encode the Hall pairing.
387. State stable-envelope transport as theorem or hypothesis.
388. Construct stable envelopes if using them.
389. Prove no toric CY3 theorem is imported without replacement.
390. Replace Schiffmann–Vasserot analogies by compact $K3 \times E$ proofs.
391. Define the Cartan part in the Drinfeld double.
392. Identify Cartan with $\Lambda_{II}^{2,1} \otimes \mathbb{C}$.
393. Prove primitive Cartan survives reduction.
394. Construct h_i^X geometrically.
395. Prove $[e_i^X, f_i^X] = h_i^X$.
396. Prove $[h_i^X, e_j^X] = (\delta_i, \delta_j) e_j^X$.
397. Prove $[h_i^X, f_j^X] = -(\delta_i, \delta_j) f_j^X$.
398. Prove $[h_i^X, h_j^X] = 0$.
399. Prove Hall bracket realizes real Serre exponent three.
400. Prove $(\text{ad } e_i^X)^3 e_j^X = 0$.
401. Prove $(\text{ad } f_i^X)^3 f_j^X = 0$.
402. Prove $[e_i^X, e_j^X] \neq 0$ for $i \neq j$.
403. Construct $E_{ij}^X = [e_i^X, e_j^X]$.
404. Prove $E_{ij}^X = -E_{ji}^X$.
405. Compute $[f_i^X, E_{ij}^X] = 2e_j^X$.
406. Compute $[f_j^X, E_{ij}^X] = -2e_i^X$.
407. Compute $[f_k^X, E_{ij}^X] = 0$.
408. Construct source isotropic directions $u_{ij,r}^X$.
409. Prove exactly nine extra isotropic simple directions.
410. Prove a_{ij} source parity is $10|0$.
411. Prove $[e_i^X, u_{ij,r}^X] = 0$.
412. Prove $[e_j^X, u_{ij,r}^X] = 0$.
413. Prove $[f_i^X, u_{ij,r}^X] = 0$.
414. Prove $[f_j^X, u_{ij,r}^X] = 0$.
415. Prove $(\text{ad } e_k^X)^5 u_{ij,r}^X = 0$.
416. Construct T_1^X, T_2^X, T_3^X .
417. Prove $T_1^X + T_2^X + T_3^X = 0$.
418. Construct the 27 mixed real-isotropic source directions.
419. Construct the 93 odd δ_{123} directions.
420. Prove source parity $29|93$ at δ_{123} .

H. Pfaffian and wall normal form

421. Define the first-order Dirac block as an actual complex.
422. Define its domain and codomain bundles.
423. Prove it is skew-adjoint or Pfaffian-admissible.
424. Define the Pfaffian line $L_{\text{Pf},R}$.
425. Prove $L_{\text{Pf},R}^{\otimes 2} \cong \det D_R$.

426. Construct finite Pfaffian section $\text{pf}_{X,R}$.
427. Prove $\text{pf}_{X,R}$ is nonzero on the complement of walls.
428. Compute its vanishing order on H_{δ_i} .
429. Prove the order is one for each type-II wall.
430. Define local coordinate u normal to a wall.
431. Prove local model is $[\mathbb{R}^u \ \mathbb{R}]$.
432. Compute the local Pfaffian sign crossing.
433. Prove sign equals -1 .
434. Prove $\epsilon_o(s_{\delta_i}) = -1$ from wall model.
435. Construct the wall atlas $O2_{\text{atlas}}$.
436. Prove wall atlas covers all type-II simple walls.
437. Prove wall atlas transports across $W^{(2)}$.
438. Prove wall atlas does not include type-I walls.
439. Define the half-Hilbert orbit.
440. Prove its size is 2^{12} .
441. Prove $2^{12} = 64^2$ is only scalar equality.
442. Prove half-Hilbert count does not define ν_{Δ_5} .
443. Prove OP minus sign does not define orientation.
444. Prove Pfaffian leading term is $64q^{1/2}r^{1/2}s^{1/2}$.
445. Compute the Pfaffian product exponents.
446. Prove exponents equal $f(nm, l)$.
447. Prove active-support coverage of Pfaffian section.
448. Prove inactive support has exponent zero.
449. Prove finite Pfaffian sections are compatible under R .
450. Prove ML exactness for Pfaffian lines.
451. Prove the inverse-limit Pfaffian section exists.
452. Prove it is holomorphic.
453. Prove it transforms with ν_{Δ_5} .
454. Prove it has weight five.
455. Prove its divisor is the type-II Humbert divisor.
456. Prove the leading coefficient fixes equality with Δ_5 .
457. State q -expansion principle used for equality.
458. Prove Pfaffian comparison is not merely determinant comparison.
459. Prove scalar trace of inverse square yields Δ_5^{-2} .
460. Prove orientation is lost under squaring.
461. Define $D0$ -Pfaffian degeneration.
462. Prove Pfaffian line extends over $D0$ limit.
463. Prove section extends over $D0$ limit.
464. Prove orientation extends over $D0$ limit.
465. Prove wall atlas extends over $D0$ limit.
466. Compute first Fourier coefficient from Pfaffian geometry.
467. Compute $f(0, 0) = 10$ from source geometry.
468. Compute $f(0, \pm 1) = 1$ from wall geometry.
469. Compute $f(1, 1) = -64$ from source geometry.
470. Compute $29|93$ before taking Pfaffian determinant.

I. Chiral E_3 , factorization, and Koszul duality

471. Define the finite holomorphic E_3 -prefactorization model.
472. State the operad acting on finite stages.
473. Prove BV quantum master equation.
474. Prove anomaly cancellation on $K3 \times E$.
475. Prove locality for projection-finite supports.
476. Prove wrapped locality is replaced by hybrid correspondences.
477. Define the $K3 \rightarrow E$ specialization functor.
478. Prove it respects vanishing cycles.
479. Prove it respects orientations.

480. Prove it respects Hall product.
481. Prove it respects E_3 operations.
482. Define $F_{X,R}^{\text{hyb}}$ as an algebra object.
483. Define its augmentation $\epsilon_{X,R}^{\text{hyb}}$.
484. Prove augmentation compatibility with hybrid product.
485. Define $\bar{B}_E^{\text{ch}}(F_{X,R}^{\text{hyb}})$.
486. Prove conilpotence of the bar coalgebra.
487. Define bar-length filtration.
488. Prove bar-length filtration is bounded at finite R .
489. Define collision coproduct Δ_R^{ch} .
490. Prove chiral coassociativity.
491. Prove counitality.
492. Prove compatibility with higher-colored coherences.
493. Define the wrapped bar-cobar domain.
494. Prove local summands lie in the domain.
495. Prove wrapped summands lie in the domain.
496. Prove mixed words lie in the domain.
497. Prove completed/coderived category is specified.
498. Prove source-internal counit is a quasi-isomorphism.
499. Define $\vartheta_R : F_{X,R}^{\text{hyb}} \rightarrow \text{Den}_{\Delta_5, E, \leq R}$.
500. Prove ϑ_R preserves differential.
501. Prove ϑ_R preserves product.
502. Prove ϑ_R preserves unit.
503. Prove ϑ_R preserves primitive restriction.
504. Prove ϑ_R preserves Gram grading.
505. Prove ϑ_R preserves parity.
506. Prove ϑ_R preserves transitions.
507. Set $\Theta_{\text{Kos}, R} = \vartheta_R \circ \epsilon_{\text{bar}, R}^{\text{src}}$.
508. Prove $\Theta_{\text{Kos}, R}$ is a quasi-isomorphism.
509. Prove its cone has ML-acyclic inverse system.
510. Prove $\Omega_E^{\text{ch}} C_X \simeq \text{Den}_{\Delta_5, E}$.
511. Prove target-internal counit is not used as source data.
512. Define $\text{Cur}_E(\mathfrak{g}_{\Delta_5})$.
513. Define $U_E^{\text{ch}}(\text{Cur}_E \mathfrak{g}_{\Delta_5})$.
514. Prove level-zero current envelope has constant Lie algebra $\mathfrak{g}_{\Delta_5}$.
515. Prove chiral envelope is compatible with completion.
516. Prove chiral envelope is compatible with super signs.
517. Prove Koszul comparison respects Weyl action.
518. Prove Koszul comparison respects Pfaffian orientation.
519. Prove Koszul comparison respects Hall pairing.
520. Prove Koszul comparison respects radical quotient.
521. Define the Beilinson–Drinfeld chiral category used.
522. Define the Francis–Gaitsgory Koszul theorem invoked.
523. Verify its hypotheses in the hybrid wrapped setting.
524. Replace “Koszul source datum” by an explicit theorem.
525. Replace “wrapped Koszul defect” by a computed obstruction group.
526. Compute that obstruction group in first window.
527. Prove the obstruction group vanishes in first window.
528. Define the homotopy inverse if claiming equivalence.
529. Prove homotopy coherence of inverse.
530. Prove chiral Koszul comparison survives $R \rightarrow \infty$.

J. Primitive recognition

531. Define $B_{\Delta_5} = (H, I, p, \alpha_i, (-, -))$ completely.
532. Define the completed free Lie superalgebra $\mathfrak{F}(B_{\Delta_5})$.
533. Define J_{BK} .

534. Define Rad_{GN} .
535. Define $G(B_{\Delta_5}) = \mathfrak{F}/(J_{BK} + Rad_{GN})$.
536. Prove $G(B_{\Delta_5}) \cong \mathfrak{g}_{\Delta_5}$.
537. Define source primitive $P_X^{\Pi, \text{red}}$.
538. Construct the Cartan map ι_H .
539. Prove ι_H is even.
540. Prove ι_H identifies degree zero with $\Lambda_H^{2,1} \otimes \mathbb{C}$.
541. Construct e_i^X, f_i^X, h_i^X for real roots.
542. Construct imaginary simple representatives.
543. Assign parity to each imaginary representative.
544. Prove GN multiplicity fibres $\mu_0(a), \mu_1(a)$ geometrically.
545. Construct source presentation map π .
546. Prove π is continuous.
547. Prove π preserves Gram grading.
548. Prove π preserves parity.
549. Prove π preserves Cartan action.
550. Prove Cartan relations in source.
551. Prove Chevalley relations in source.
552. Prove real Serre relations in source.
553. Prove Borchers orthogonality in source.
554. Prove super sign conventions in source.
555. Prove isotropic-string relations in source.
556. Prove $J_{BK} \subset \ker \pi$.
557. Compute $\ker \pi$ in $W_{\leq 1}$.
558. Compute $\ker \pi$ in $W_{\leq 2}$.
559. Compute $\ker \pi$ in $W_{\leq 3}$.
560. Compute $\ker \pi$ in first relation-closed window.
561. Prove $\ker \pi = J_{BK} + Rad_{GN}$.
562. Prove source radical equals GN radical.
563. Prove source radical is closed.
564. Prove source radical is homogeneous.
565. Prove source radical is a Lie ideal.
566. Prove source radical is a coideal.
567. Prove quotient pairing is nondegenerate.
568. Prove π descends to the radical quotient.
569. Prove descended π is injective.
570. Prove descended π is surjective.
571. Prove source generation by simple representatives.
572. Prove no extra simple source generators appear.
573. Prove every source primitive is a bracket word.
574. Prove every bracket word has target counterpart.
575. Prove PBW filtration exists on source enveloping algebra.
576. Prove PBW associated graded matches target.
577. Prove PBW compatibility on every finite window.
578. Prove PBW transitions are strict.
579. Prove exact triangular completion.
580. Prove no positive-negative cancellation generators are added.
581. Prove $\lim^1$ vanishes for kernels.
582. Prove $\lim^1$ vanishes for radicals.
583. Prove $\lim^1$ vanishes for PBW gradeds.
584. Prove $\lim^1$ vanishes for parity pieces.
585. Compute full parity at δ_i .
586. Compute full parity at a_{ij} .
587. Compute full parity at $2\delta_i + \delta_j$.
588. Compute full parity at δ_{123} .

589. Compute full parity at $2a_{ij}$.
590. Compute full parity at $C'_{k,2}$.
591. Compute full parity at $C'_{k,3}$.
592. Compute full parity at $C'_{k,4}$.
593. Compute full parity at $C'_{k,5}$.
594. Compute full parity at $2\delta_{123}$.
595. Mark signed-only rows unusable for recognition.
596. Build the first relation-closed target window.
597. Build the corresponding source window.
598. Compute source bases in that window.
599. Compute product matrices M .
600. Compute coproduct matrices D .
601. Compute bracket matrices B .
602. Compute pairing matrices G .
603. Compute kernel matrices K .
604. Compute quotient matrices Q .
605. Compute comparison maps $A_{\beta,p}$.
606. Verify finite relation identities.
607. Verify no-extra kernel equality.
608. Verify PBW associated graded equality.
609. Verify strict ML transition maps.
610. Prove negative-root parity via Chevalley anti-involution.
611. Prove source Chevalley anti-involution exists.
612. Prove it preserves orientation.
613. Prove it preserves pairing.
614. Prove it preserves radical.
615. Prove it preserves PBW.
616. Prove it matches target anti-involution.
617. Rule out hidden $1|1$ cancelling source summands.
618. Rule out wrong parity lifts invisible to denominator.
619. Rule out parity-cancelling ideals.
620. Rule out zero-bracket denominator model.
621. Prove signed multiplicities follow from full parity.
622. Prove denominator identity follows after recognition.
623. Prove recognition does not follow from denominator.
624. Prove recognition does not follow from Pfaffian.
625. Prove recognition does not follow from scalar trace.
626. Prove recognition uses orientation through Frobenius pairing.
627. Prove recognition uses Koszul comparison through source coalgebra.
628. Prove recognition uses Hall bialgebra through primitives.
629. Prove recognition uses normal-ordering through Gram descent.
630. Prove recognition survives inverse limit.

K. Mirror period side

631. Define the rank-3 Mukai–Hodge quotient $P_{K3 \times E}^{(3)}$.
632. Construct it as an actual period domain quotient.
633. Define the period Hodge line λ_{per} .
634. Construct the comparison $L_{\text{aut}} \cong \lambda_{\text{per}}$.
635. Prove comparison on the open complement.
636. Prove extension across the discriminant.
637. Define the mirror discriminant divisor $\mathcal{D}_{\text{mir}}$.
638. Prove its support is the type-II Heegner divisor.
639. Prove multiplicity one for $\mathcal{D}_{\text{mir}}$.
640. Prove Δ_5^{-2} has polar order two.
641. Prove $\text{div}_{\infty}(\Delta_5^{-2}) = 2\mathcal{D}_{\text{mir}}$.
642. Compute local Picard–Lefschetz monodromy.

- 643. Compute Hodge-line exponent from Picard–Lefschetz.
- 644. Compare with automorphic weight ten.
- 645. Prove the period quotient is actually rank three.
- 646. Separate U_{mir} from the elliptic even lattice.
- 647. Prove the $PSp_4 \simeq SO(3, 2)^+$ bridge at the lattice level.
- 648. Prove sign convention $2U \oplus \langle -2 \rangle \cong \Lambda^{3,2}$.
- 649. Prove type-II Heegner orbit corresponds to Humbert surface.
- 650. Prove irreducibility of the divisor in the quotient.
- 651. Prove no type-I divisor contributes to the mirror discriminant.
- 652. Prove the conjecture does not imply primitive recognition.
- 653. Prove primitive recognition does not imply the conjecture.
- 654. Prove OP scalar identity gives only trace-level evidence.
- 655. Remove mirror material from proof of Pfaffian theorem.
- 656. Define VHS explicitly.
- 657. Define monodromy group explicitly.
- 658. Compute its arithmetic image.
- 659. Compute discriminant multiplicities from local degenerations.
- 660. Compare discriminant multiplicities with Δ_5^{-2} .

L. Scalar trace, BPS, and physics-mathematics interface

- 661. Define $Z_{K3 \times E}^{\text{BPS}}$ as a mathematical trace.
- 662. Define the trace category.
- 663. Define the operator being traced.
- 664. Prove the trace is Δ_5^{-2} .
- 665. Prove OP normalization gives $-4096\Delta_5^{-2}$.
- 666. Prove the minus sign is scalar branch only.
- 667. Prove 4096 is squared theta-leading coefficient.
- 668. Prove 4096 is also half-Hilbert count only separately.
- 669. Prove equality of numbers does not identify structures.
- 670. Define the protected trace functor.
- 671. Prove trace kills ν_{Δ_5} .
- 672. Prove trace forgets orientation character.
- 673. Prove inverse square cannot recover Pfaffian sign.
- 674. Prove scalar trace cannot recover bracket.
- 675. Prove scalar trace cannot recover parity split.
- 676. Prove scalar trace cannot recover Hopf pairing.
- 677. Prove scalar trace cannot recover PBW.
- 678. Define any physical BPS index as conjectural unless proved.
- 679. Separate CHL dyon-counting from $K3 \times E$ compact source.
- 680. Separate $K3 \times T^2$ physics from $K3 \times E$ algebraic geometry.
- 681. Define what “protected” means mathematically.
- 682. Define the cohomological grading of protection.
- 683. Define fermion parity $(-1)^F$.
- 684. Prove parity agrees with BKM super parity.
- 685. Prove parity is compatible with orientation.
- 686. Prove parity is compatible with Hall product.
- 687. Prove parity is compatible with Koszul differential.
- 688. Prove parity is compatible with Weyl transport.
- 689. State gravity-line algebra as open unless constructed.
- 690. Do not use path-integral language as evidence.

M. References, internal consistency, and normalization discipline

- 691. Verify every theorem citation actually proves the stated claim.
- 692. Separate Borchers 1995 from Borchers 1998 statements.
- 693. Separate Gritsenko–Nikulin denominator theorem from additive lift theorem.
- 694. Separate Igusa theta-constant theorem from Borchers product theorem.

- 695. Separate Oberdieck–Pixton from Oberdieck–Pandharipande normalizations.
- 696. Verify Maass multiplier formula in your conventions.
- 697. Verify $Sp(2, \mathbb{Z})$ means $Sp_4(\mathbb{Z})$.
- 698. Use one notation for Siegel upper half-space.
- 699. Use one notation for Δ_5, D_5, D_X .
- 700. Use one notation for ϵ_o or ε_o .
- 701. Use one notation for $\mathcal{D}_X^{DI}$.
- 702. Use one notation for P_X .
- 703. Use one notation for Π_X quadratic map.
- 704. Use different notation for normal-ordered Π_X .
- 705. Avoid using Π_X for two maps without warning.
- 706. Distinguish Γ_{gram} from Γ_{eff} .
- 707. Distinguish Γ_X^{eff} from Γ_{eff} .
- 708. Distinguish geometric b from Borchers m .
- 709. Distinguish $m(a)$ correction coefficients from coordinate m .
- 710. Distinguish $\tau(a)$ isotropic exponent from transport τ_i .
- 711. Distinguish $D0$ degeneration from D0-brane charge.
- 712. Distinguish $K3$ topological Euler 24 from $\chi(O_{K3}) = 2$.
- 713. Distinguish κ_{Heis} from κ_{BKM} .
- 714. Distinguish κ_{fiber} from BKM weight.
- 715. Prove $\kappa_{\text{BKM}} = c_1(0)/2 = 5$.
- 716. Do not use accidental $3 + 2 = 5$ as structure.
- 717. Distinguish paramodular N from CHL twining order.
- 718. Verify CHL constant table separately.
- 719. Verify eight-row Cléry–Gritsenko data separately.
- 720. Mark half-integral weight rows carefully.
- 721. State every lattice signature convention before use.
- 722. State whether signature means $(+, -)$ or $(-, +)$.
- 723. Verify $\Lambda^{3,2} \cong 2U \oplus \langle 2 \rangle$ in chosen sign.
- 724. Verify $\Lambda^{2,1} \cong U \oplus \langle 2 \rangle$.
- 725. Verify $\Lambda_{II}^{2,1} \cong U(4) \oplus \langle 2 \rangle$.
- 726. Verify dual lattice computations.
- 727. Verify discriminant group order 32.
- 728. Verify type-I/type-II divisibility statements.
- 729. Verify $O^+(\Lambda^{2,1}) = W^{(2)}(\Lambda^{2,1})$ citation.
- 730. Verify index statements for reflection subgroups.

N. Concrete first-window computations

- 731. Construct source class for δ_1 .
- 732. Construct source class for δ_2 .
- 733. Construct source class for δ_3 .
- 734. Compute moduli stack for δ_1 .
- 735. Compute moduli stack for δ_2 .
- 736. Compute moduli stack for δ_3 .
- 737. Compute H^{BM} for δ_1 .
- 738. Compute H^{BM} for δ_2 .
- 739. Compute H^{BM} for δ_3 .
- 740. Show each gives $1|0$.
- 741. Construct a_{12} source stack.
- 742. Construct a_{13} source stack.
- 743. Construct a_{23} source stack.
- 744. Compute real bracket direction in a_{ij} .
- 745. Compute nine extra isotropic directions geometrically.
- 746. Show a_{ij} gives $10|0$.
- 747. Construct $2\delta_1 + \delta_2$ source stack.
- 748. Construct $2\delta_1 + \delta_3$ source stack.

- 749. Construct $2\delta_2 + \delta_1$ source stack.
- 750. Construct $2\delta_2 + \delta_3$ source stack.
- 751. Construct $2\delta_3 + \delta_1$ source stack.
- 752. Construct $2\delta_3 + \delta_2$ source stack.
- 753. Show each real string degree gives $1|0$.
- 754. Construct δ_{123} source stack.
- 755. Compute its even part 29.
- 756. Compute its odd part 93.
- 757. Verify $29 - 93 = -64$.
- 758. Construct source T_i brackets.
- 759. Verify $T_1 + T_2 + T_3 = 0$.
- 760. Construct $M_{ij,r}$ mixed brackets.
- 761. Verify the 27 mixed brackets are independent.
- 762. Construct 93 odd generators as vanishing-cycle classes.
- 763. Verify their pairing matrix.
- 764. Verify their brackets with real generators.
- 765. Verify no additional δ_{123} source classes survive.
- 766. Compute source $2a_{ij}$ parity.
- 767. Compute source $C'_{k,2}$ parity.
- 768. Compute source $C'_{k,3}$ parity.
- 769. Compute source $C'_{k,4}$ parity.
- 770. Verify terminal zero $C_{k,5}$.
- 771. Compute source $2\delta_{123}$ parity.
- 772. Do not use signed-only target rows as parity data.
- 773. Extend target presentation to signed-only rows.
- 774. Then compare source parity to extended target parity.
- 775. Compute first PBW graded dimensions.
- 776. Compute first Hall radical kernels.
- 777. Compute first kernel equality.
- 778. Compute first relation-closed window fully.
- 779. Publish the matrices in an appendix.
- 780. Make the first-window theorem self-contained.

O. Spectral sequence and homotopical edifice

- 781. Define the Hall filtration by number of simple factors.
- 782. Define the associated graded correspondence complex.
- 783. Define the flag complex resolving primitives.
- 784. Construct the Hall–Chevalley complex.
- 785. Identify E^1 with free Lie words in simple source classes.
- 786. Identify d_1 with Hall boundary.
- 787. Prove d_1 imposes Chevalley relations.
- 788. Prove d_1 imposes Serre relations.
- 789. Prove d_1 imposes orthogonality relations.
- 790. Prove higher differentials vanish in first window.
- 791. Compute E^2 in first window.
- 792. Compare E^2 with GN target presentation.
- 793. Prove spectral sequence converges strongly.
- 794. Prove convergence is compatible with $R \rightarrow \infty$.
- 795. Prove no hidden extensions alter parity.
- 796. Use Frobenius pairing to kill hidden radical classes.
- 797. Use PBW to compare associated gradeds.
- 798. Define obstruction groups for higher differentials.
- 799. Compute obstruction groups in low degree.
- 800. Prove obstruction vanishing recursively or state precise conjecture.
- 801. Define the cohomological Hall operad action.
- 802. Relate the operad action to E_3 .

- 803. Relate E_3 Koszul duality to chiral Koszul duality on E .
- 804. Prove dimensional reduction from $K3 \times E$ to E .
- 805. Prove reduced K3 direction contributes the correct Frobenius degree.
- 806. Prove the spectral sequence respects Weyl reflections.
- 807. Prove Weyl reflections act on pages.
- 808. Prove Weyl action on E_∞ gives Maass character.
- 809. Prove Pfaffian line is determinant of the spectral complex.
- 810. Prove determinant of spectral complex gives Borchers product.

P. Boundary, degenerations, and missing compactness

- 811. Define the $D0$ -degeneration family.
- 812. Prove the central fibre exists.
- 813. Prove reduced obstruction theory specializes.
- 814. Prove HN filtrations specialize.
- 815. Prove orientations specialize.
- 816. Prove Hall products specialize.
- 817. Prove Pfaffian sections specialize.
- 818. Prove Koszul coalgebras specialize.
- 819. Prove primitive recognition specializes.
- 820. Prove no boundary component contributes extra roots.
- 821. Treat singular K3 limits or explicitly exclude them.
- 822. Treat nodal elliptic degenerations or explicitly exclude them.
- 823. Treat cuspidal elliptic degenerations or explicitly exclude them.
- 824. Treat imprimitive curve classes or explicitly exclude them.
- 825. Treat $s = 0$ compact Hall cusp or explicitly exclude it.
- 826. Treat multicomponent charge closure or explicitly exclude it.
- 827. Prove exclusions are stable under Hall product.
- 828. Prove exclusions are stable under coproduct.
- 829. Prove exclusions are stable under Weyl reflection.
- 830. Prove exclusions are stable under R -transition.

Q. Line-by-line theorem upgrades

- 831. Turn Definition 1.1 into a theorem only after View III is proved.
- 832. Turn Pfaffian–Dirac statement into “relative theorem” until $D0, O1, O1^+, O2$ are computed.
- 833. Add proof of P_{fin} .
- 834. Add proof of $D0$ -HN datum.
- 835. Add proof of quotient-orientation datum.
- 836. Add proof of Weyl-lift datum.
- 837. Add proof of $O2_{\text{atlas}}$.
- 838. Add proof of E_3 -HolFA datum.
- 839. Add proof of $K3 \rightarrow E$ specialization.
- 840. Add proof of chiral Koszul source datum.
- 841. Add proof of Koszul ML exactness.
- 842. Add proof of cofinal recognition datum.
- 843. Replace “standing hypothesis” by proposition or axiom label.
- 844. Replace “retained residual vanishes” by cohomology-class equation.
- 845. Replace “compatible” by commutative diagram.
- 846. Replace “finite-stage” by explicit finite category.
- 847. Replace “source representative” by a cycle class.
- 848. Replace “pairing matrix” by an actual matrix.
- 849. Replace “kernel equality” by computed kernel basis.
- 850. Replace “PBW comparison” by associated graded isomorphism.
- 851. Replace “full parity” by two integer equations.
- 852. Replace “orientation character” by homomorphism from Weyl group.
- 853. Replace “wall sign” by local determinant calculation.
- 854. Replace “Pfaffian section” by determinant of complex.

- 855. Replace "protected trace" by categorical trace formula.
- 856. Replace "state space" by constructed cohomology object.
- 857. Replace "operator algebra" by chiral algebra object.
- 858. Replace "bulk" by derived chiral center if used.
- 859. Replace "bar coalgebra" by MC twisting coalgebra.
- 860. Replace "gravity line" by open problem unless constructed.

R. Things to remove from theorem proofs unless proved

- 861. Do not use mirror discriminant in Pfaffian proof.
- 862. Do not use OP scalar branch in orientation proof.
- 863. Do not use Δ_5^{-2} in primitive recognition.
- 864. Do not use signed dimensions as parity dimensions.
- 865. Do not use target labels as source representatives.
- 866. Do not use target current envelope as source coalgebra.
- 867. Do not use target counit as source Koszul map.
- 868. Do not use toric CoHA theorems for compact non-toric $K3 \times E$ without replacement.
- 869. Do not use quiver-with-potential bialgebra theorem without compact analogue.
- 870. Do not use physical duality as proof of Hall equivalence.
- 871. Do not use $K3 \times T^2$ charges as $K3 \times E$ sheaf charges.
- 872. Do not infer full Weyl action from three generators without domain.
- 873. Do not infer semidirect S_3 lift from type-II Weyl lift.
- 874. Do not infer finite moduli from formal Mukai lifts.
- 875. Do not infer effectivity from formal lattice arithmetic.
- 876. Do not infer algebraicity from full Mukai lattice existence.
- 877. Do not infer compactness from projectivity of X .
- 878. Do not infer properness of extension correspondences automatically.
- 879. Do not infer coproduct from product by reversal.
- 880. Do not infer Hopf pairing from Poincaré duality.

S. Minimal publishable core if everything is upgraded

- 881. Prove all classical BGN/GN/Borcherds statements cleanly.
- 882. Make the $K3 \times E$ part a precise conditional theorem.
- 883. Prove one finite relation-closed source window.
- 884. Include all matrices for that window.
- 885. Prove $1|0, 10|0, 1|0, 29|93$ source parity.
- 886. Prove Chevalley and Serre relations in that window.
- 887. Prove radical coideal property in that window.
- 888. Prove no-extra kernel equality in that window.
- 889. Prove PBW comparison in that window.
- 890. Prove Pfaffian wall sign for the three simple walls.
- 891. Prove orientation character on the three simple walls.
- 892. Prove $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}$.
- 893. Prove finite Pfaffian leading coefficient.
- 894. Prove $\text{Pf}^{\text{prot}} = \Delta_5$ relative to constructed finite data.
- 895. Leave global primitive recognition as conjecture if not completed.
- 896. Leave mirror discriminant as conjecture.
- 897. Leave gravity-line algebra as conjecture.
- 898. Leave compact-source all-window recognition as conjecture unless proved.
- 899. Do not call conjectural parts theorem-level.
- 900. Make the first-window theorem the mathematical centerpiece.

T. The deepest algebraic endpoint

- 901. Construct $P_X^{\text{II}, \text{red}}$ as a derived Lie algebra, not just H^0 .
- 902. Identify its L_∞ minimal model.
- 903. Prove higher L_∞ operations vanish or match a BKM enhancement.
- 904. Define derived BKM enhancement of $\mathfrak{g}_{\Delta_5}$.

- 905. Prove strictification from L_∞ to Lie superalgebra.
- 906. Prove obstruction classes to strictification vanish.
- 907. Define E_3 -Koszul dual of the Hall source.
- 908. Compare with chiral enveloping algebra on E .
- 909. Prove Dunn additivity in this hybrid setting.
- 910. Prove factorization homology over $K3$ gives the E -chiral source.
- 911. Compute the Hochschild/chiral center level.
- 912. Identify $Z_{\text{ch}}^{\text{der}}$ if claimed.
- 913. Prove bar coalgebra is not the center.
- 914. Construct the center separately if needed.
- 915. Define the CY3 trace pairing categorically.
- 916. Prove it descends through cosection.
- 917. Prove it induces the Hall Frobenius pairing.
- 918. Prove it induces Hopf radical coideal stability.
- 919. Prove it induces primitive recognition pairing.
- 920. Prove it induces Pfaffian orientation.

U. Final normalization and consistency checks

- 921. Check every exponent $f(nm, l)$ against $\phi_{0,1}$.
- 922. Check every root degree against $\alpha(n, l, m)$.
- 923. Check every finite window is downward saturated.
- 924. Check every relation window is relation closed.
- 925. Check every source degree has target image.
- 926. Check every target root has source representative.
- 927. Check every source representative has compact provenance.
- 928. Check every compact class is algebraic.
- 929. Check every algebraic class is effective only when proved.
- 930. Check every effective class has nonempty moduli only when proved.
- 931. Check every moduli stack has finite-type proof.
- 932. Check every correspondence has proper/admissible leg proof.
- 933. Check every six-functor operation has category specified.
- 934. Check every orientation line has square-root proof.
- 935. Check every gerbe class has vanishing proof.
- 936. Check every Weyl lift has coherence proof.
- 937. Check every Pfaffian sign has local model proof.
- 938. Check every Koszul equivalence has cone-acyclicity proof.
- 939. Check every recognition equality has matrix proof.
- 940. Check every inverse limit has ML proof.

V. The mathematical "platonic ideal" sentences

- 941. State: "The determinant is not recognition."
- 942. State: "The scalar trace is not orientation."
- 943. State: "The target BKM algebra is not the source Hall algebra."
- 944. State: "The source-to-target map is a theorem, not a counit."
- 945. State: "The compact source data are the main construction."
- 946. State: "The first relation-closed finite window is the first test."
- 947. State: "Full parity, not signed dimension, is required."
- 948. State: "The Hopf radical is structural, not cosmetic."
- 949. State: "PBW is the algebraic spine of recognition."
- 950. State: "The Pfaffian line must be built before it equals Δ_5 ."

W. Last 50 high-impact repairs

- 951. Give a theorem proving $O1$ in the first window.
- 952. Give a theorem proving $O1^+$ in the first window.
- 953. Give a theorem proving $O2$ in the first window.
- 954. Give a theorem proving P_{fin} in the first window.

955. Give a theorem proving Hall product in the first window.
956. Give a theorem proving Hall coproduct in the first window.
957. Give a theorem proving Hopf pairing in the first window.
958. Give a theorem proving source Koszul comparison in the first window.
959. Give a theorem proving primitive recognition in the first window.
960. Give a theorem proving Pfaffian equality in the first window.
961. Compute e_i^X geometrically.
962. Compute f_i^X geometrically.
963. Compute h_i^X geometrically.
964. Compute $u_{ij,r}^X$ geometrically.
965. Compute odd δ_{123} generators geometrically.
966. Compute the first source bracket table.
967. Compute the first source coproduct table.
968. Compute the first source pairing table.
969. Compute the first radical quotient table.
970. Compute the first PBW table.
971. Compare all five tables to GN target tables.
972. Show the zero-bracket counterexample fails exactly at $(S4), (S5), (S6)$.
973. Show any wrong parity lift fails $(S10)$.
974. Show any extra source generator fails $(S6)$ or $(S8)$.
975. Show any radical mismatch fails Hopf coideal stability.
976. Show any nonproper correspondence breaks coproduct.
977. Show any nontrivial stabilizer character breaks orientation descent.
978. Show any target-counit shortcut breaks source-target separation.
979. Show any scalar-trace shortcut breaks orientation.
980. Show any mirror shortcut breaks logical independence.
981. Prove the strongest unconditional theorem: Views I, II, V.
982. Prove the strongest relative theorem: Pfaffian/orientation under constructed $D0, O1, O1^+, O2, P_{\text{fin}}$.
983. Prove the strongest finite theorem: first-window primitive recognition.
984. State the strongest conjecture: all-window compact-source recognition.
985. State the strongest period conjecture: $\Delta_5^{-2} = \text{Disc}_{\text{mir}}$.
986. State the strongest homotopy conjecture: $\Omega_E^{ch} C_X \simeq \text{Den}_{\Delta_5, E}$.
987. State the strongest Hall conjecture: compact non-toric CY3 Drinfeld double.
988. State the strongest Pfaffian theorem as a line-bundle equality.
989. State the strongest BKM theorem as a Lie-superalgebra isomorphism.
990. State the strongest physics theorem only after constructing a state space.
991. Put all obstruction groups in one appendix.
992. Put all finite matrices in one appendix.
993. Put all lattice computations in one appendix.
994. Put all normalizations in one appendix.
995. Put all target BKM parity computations in one appendix.
996. Put all source geometric computations in one appendix.
997. Put all ML exactness proofs in one appendix.
998. Put all Koszul duality hypotheses in one appendix.
999. Put all mirror-period conjectures in a separate section.
1000. Make the first relation-closed compact-source recognition theorem the paper's central mathematical jewel.

The single most important repair is not stylistic. It is this:

Construct and compute the source primitive Hall Lie superalgebra in one relation-closed window.

Until then, the manuscript is a powerful conditional atlas around Δ_5 , not yet the $K3 \times E$ realization theorem it wants to be.

       Sources

